The following text is from Darcie Little Badger's 2021 novel *A Snake Falls to Earth*. Nina is looking at old family photographs.

> Nina couldn't stop looking at the sepia-toned photograph in an oak frame. Propped on the window ledge, it featured a portrait of Great-Great-Grandmother Rosita as a young woman.

©2021 by Darcie Little Badger

1 Mark for Review

As used in the text, what does the word "featured" most nearly mean?

(A) Ignored

(B) Approved

(C) Questioned

(D) Displayed

@DSATUZ

The following text is from Mark Twain's 1876 novel *The Adventures of Tom Sawyer*. Tom, a child, has been told by his aunt to paint their house's fence.

> Tom appeared on the sidewalk with a bucket of whitewash and a long-handled brush. He surveyed the fence, and all gladness left him and a deep melancholy settled down upon his spirit. Thirty yards of board fence nine feet high.

2

As used in the text, what does the word "surveyed" most nearly mean?

(A) Promoted

(B) Organized

(C) Adapted

(D) Examined

@DSATUZ

Politicians routinely claim that journalists are _____, but while it's true that some journalists allow their own views to creep into their reporting more than others do, most journalists make a good-faith effort to be neutral in their reporting.

Which choice completes the text with the most logical and precise word or phrase?

(A) sensitive

(B) irrelevant

(C) biased

(D) misinformed

@DSATuz

The Danube River delta system is located in Romania, where the river drains into the Black Sea, and is shaped by _____ factors: for example, the geography of the coastline influences sedimentary deposition, which over time alters coastal geography.

Which choice completes the text with the most logical and precise word or phrase?

(A) comprehensive

(B) unyielding

(C) interdependent

(D) tenuous

@DSATUZ

The following text is adapted from Jason Reynolds's 2016 novel *Ghost*. The narrator, who is in middle school, is at a bus stop.

> I just go there [to the bus stop] to look at the people working out. See, the gym across the street has this big window—like the whole wall is a window—and they have those machines that make you feel like you walking up steps and so everybody just be facing the bus stop, looking all crazy like they're about to pass out. And trust me, there ain't nothing funnier than that. So I check that out for a little while like it's some kind of movie: *The About to Pass Out Show*, starring stair-stepper person one through ten.

©2016 by Jason Reynolds

5

Which choice best states the main purpose of the text?

(A) To discuss a movie that the narrator saw in a theater

(B) To explain a problem that the narrator has overcome

(C) To describe an activity that the narrator finds amusing

(D) To give a reason why the narrator is excited to start middle school

@DSATUZ

In 1967, Ed Bradley was hired by CBS News and eventually became the first Black White House correspondent for CBS. Bradley's accomplishment is just one example of the rich history of Black journalism in the United States. That history is preserved by the National Association of Black Journalists, which was founded in Washington, DC, in 1975 to support Black media professionals and honor people like Bradley.

6 Mark for Review

Which choice best describes the function of the underlined sentence?

(A) It summarizes the purpose of founding the National Association of Black Journalists.

(B) It highlights several important members of the National Association of Black Journalists.

(C) It emphasizes the challenges that the National Association of Black Journalists has overcome.

(D) It lists the achievements of the National Association of Black Journalists.

@DSATuz

Barn owls and cuttlefish see in three dimensions (3D) by combining two images in their brains, one from each eye. This produces a sense of depth, helping the animals judge how close or far away an object is. Researchers have investigated 3D vision in praying mantises as well. In one study, Vivek Nityananda and his team fitted mantises' faces with two different color filters, one covering each eye, much like the filters in 3D glasses once worn at movies. By observing the mantises' reaction to projected images, the team confirmed that mantises do indeed have 3D vision, but it's unlike that of other animals.

7 Mark for Review

Which choice best describes the overall structure of the text?

(A) The text reviews the similarities in how barn owls and cuttlefish see, then provides details of a study about praying mantises to confirm these similarities.

(B) The text describes how barn owls and cuttlefish see, then presents a study about praying mantises that challenges the accuracy of that description.

(C) The text explains the general differences between the vision of barn owls and cuttlefish, then introduces a study of praying mantises' vision.

(D) The text discusses 3D vision in barn owls and cuttlefish, then introduces a study about this kind of vision in praying mantises.

@DSATuz

"Tomato" is one example of an English word that has an Indigenous origin. The word came to English indirectly from *tomate*, the Spanish word for the widely cultivated plant. Spanish had borrowed it from Nahuatl, an Indigenous language of Central Mexico, in which the word's original form is *tomatl*. "Guava" also has an Indigenous origin and entered English through Spanish. But in this case, the original source was Taíno, a language of the Caribbean islands, in which the word for the widely cultivated fruit is *guayabo*.

8 Mark for Review

According to the text, what is one difference between the English words "tomato" and "guava"?

(A) "Tomato" can be traced to the Nahuatl language, while "guava" can be traced to the Taíno language.

(B) "Tomato" changed in meaning significantly when English borrowed the word from Spanish, whereas "guava" retained its original meaning.

(C) "Tomato" entered English recently, but "guava" entered English many centuries ago.

(D) "Tomato" is known for being Indigenous in origin, while "guava" is mistakenly regarded as Spanish in origin.

@DSATuz

In present-day Chiapas, Mexico, archaeologist Robert Rosenswig, remote-sensing specialist Ricardo López-Torrijos, and colleagues have located 41 smaller settlements surrounding the ancient Mesoamerican city of Izapa. The researchers have concluded that these settlements were culturally linked to Izapa because each of the settlements is the same age and configured in the same manner as Izapa, with a pyramid to the north and a plaza to the south. Their shared structural orientation suggests that residents of the settlements likely performed some of the same cultural ceremonies as residents in Izapa did.

9 Mark for Review

Which choice best states the main idea of the text?

(A) Cultural ceremonies in Izapa seem to have played a more important role for its residents than those in smaller, surrounding settlements did.

(B) Researchers have inferred that Izapa was related to the smaller settlements that surrounded it based in part on the similarity of their construction.

(C) Researchers have determined that the arrangement of Izapa's structures was based on those of other nearby settlements.

(D) Although archaeologists have learned much about Izapa over years of research, they have only recently found the smaller settlements that surrounded it.

@DSATuz

Highest-Grossing Films in a Language Other than English at US Box Office

Title	Lifetime gross earnings	Opening weekend box office gross earnings	US release date	Director	Oscar nominated?
Huevos: Little Rooster's Egg-cellent Adventure	$9,080,818	$3,424,702	September 4, 2015	Gabriel Riva Palacio Alatriste and Rodolfo Riva Palacio Alatriste	No
Cinema Paradiso	$11,990,401	$16,552	February 2, 1990	Giuseppe Tornatore	Yes
The Motorcycle Diaries	$16,781,387	$159,819	September 24, 2004	Walter Salles	Yes
Como agua para chocolate	$21,665,468	$23,600	February 19, 1993	Alfonso Arau	No

Many films in a language other than English grow to be financially successful over the course of their time in movie theaters in the United States, but some become immediate successes in their opening weekends. Such success is driven by many factors like advertising, genre popularity, and the fame of the actors and directors. A student claims that opening weekend earnings can reliably predict whether a film will be nominated for an Oscar: films that draw large audiences at the beginning of their release are the most likely contenders to earn these coveted award nominations.

10 Mark for Review

Which choice best describes data from the table that weaken the student's claim?

(A) Although both *The Motorcycle Diaries* and *Cinema Paradiso* were recognized by the Oscars, *The Motorcycle Diaries* had higher lifetime earnings.

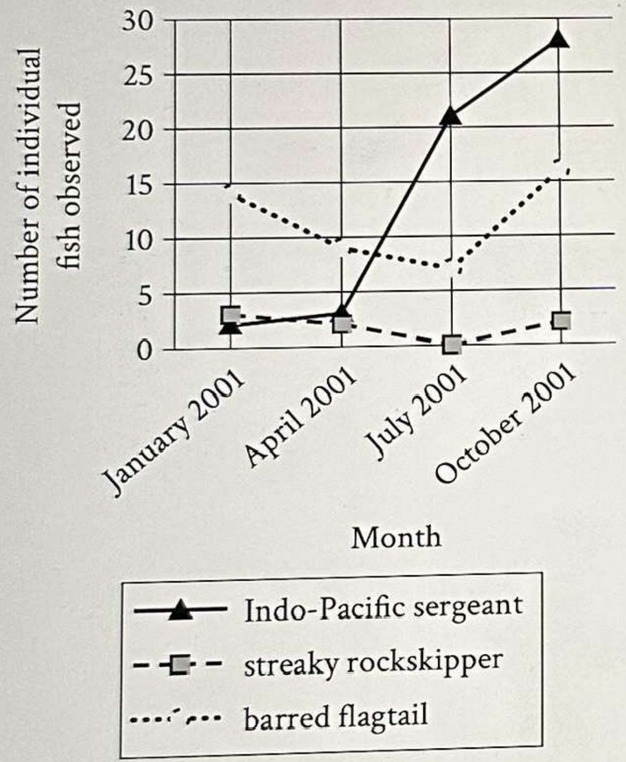
(B) Although both *Como agua para chocolate* and *Huevos: Little Rooster's Egg-cellent Adventure* were directed by well-known figures, neither film was recognized by the Oscars.

(C) *Como agua para chocolate* had higher lifetime earnings than *Cinema Paradiso* did, even though *Cinema Paradiso* was recognized by the Oscars.

(D) *Huevos: Little Rooster's Egg-cellent Adventure* was not recognized by the Oscars but had much higher opening weekend earnings than *The Motorcycle Diaries*, which was recognized by the Oscars.

@DSATuz

Fish Population in a Taiwanese Tide Pool, January 2001 to October 2001



Lin-Tai Ho and colleagues monitored fish populations in a tide pool in Taiwan. They found that there could be significant within-species variations in the number of individuals in the tide pool over time; for example, the _____

11 Mark for Review

Which choice most effectively uses data from the graph to complete the example?

(A) streaky rockskipper population was at its highest in July of 2001 and at its lowest in January of 2001.

(B) streaky rockskipper population never exceeded 3 individuals, whereas the Indo-Pacific sergeant population reached 28 individuals in October of 2001.

(C) barred flagtail population reached its highest level in October of 2001, when 16 individuals were observed.

(D) Indo-Pacific sergeant population ranged from a low of 2 individuals in January of 2001 to a high of 28 individuals in October of 2001.

@DSATuz

Numbers of the 23 Non-native Tree Species Reported and the Insect and Fungus Threats to Them

Country	Trees	Fungi	Insects
Italy	14	57	42
Ireland	7	14	6
Finland	6	11	28

Elisabeth Pötzelsberger and colleagues gathered data on 23 non-native tree species grown in Europe. They analyzed reports from Italy, Finland, and Ireland about the number of these species grown in those countries as well as the numbers of insect and fungus species that damage those trees. The researchers concluded that in only one of these countries are there more insect species than fungus species that pose risks to these trees.

@DSATuz

12 Mark for Review

Which choice best describes data from the table that support Pötzelsberger and colleagues' conclusion?

(A) Italy and Ireland both reported fewer damaging insect species than damaging fungus species, whereas Finland reported 28 fungus species and only 6 insect species.

(B) Finland reported 11 damaging fungus species and 28 damaging insect species, while both Italy and Ireland reported a greater number of damaging fungus species than insect species.

(C) Italy reported 57 damaging fungus species, whereas Ireland reported 6 damaging insect species.

(D) Italy reported 57 damaging fungus species, which is a greater number of them than either Ireland or Finland reported.

In countries with right-hand traffic, drivers who want to make a left turn at a traffic intersection with stoplights have to wait for either a gap in oncoming traffic or a designated left-turn signal to turn green. At busy intersections, this often causes a backup of vehicles waiting to turn left or being prevented from proceeding by left-turning vehicles in front of them. Transportation researcher Vikash V. Gayah claims that in urban areas eliminating the option to turn left at busy intersections—both with and without dedicated left-turn signals—would improve traffic flow and, as a result, reduce overall travel times even if such a restriction would require drivers to sometimes travel a slightly longer distance.

13 Mark for Review

Which finding, if true, would most directly support the researcher's claim?

(A) Statistics reveal that school buses in a city that eliminated left turns at most intersections took on average two minutes longer to complete their routes after the restriction took effect than they did before.

(B) In a town that installed left-turn signals at all busy intersections, seven out of ten survey respondents agreed with the statement "the streets in my community are easier to navigate by motor vehicle than before."

(C) After a city eliminated left turns at busy intersections, a package-delivery company reports that its drivers have been able to reach more addresses in the city daily, on average, and therefore deliver more packages there annually.

(D) A traffic study of intersections in a large city shows that on average drivers wait longer to make a left turn at intersections without left-turn signals than at intersections with such signals.

@DSATuz

Days per Winter That Lakes Have Surface Ice

Lake	Latitude (degrees)	1980–81	1985–86	1990–91	1995–96	2000–01	2005–06
Lake Baikal	51.85	96	118	92	103	127	109
Lake Monona	43.05	86	113	88	107	108	94
Kitusjärvi	62.283	170	176	169	188	127	149

It is common for freshwater lakes near or above a latitude of 45° north of the equator, like Lake Mjøsa in Norway, to accumulate surface ice in winter. A study from 1980 to 2006 showed that, in general, the number of days per winter that such lakes have measurable amounts of surface ice is declining. However, a researcher claimed that some lakes have instead seen an increase in the duration of ice.

@DSATUZ

14 Mark for Review

Which choice best describes data in the table that support the researcher's claim?

(A) Kitusjärvi had more days of ice in the winter of 1980–81 than it did in the winter of 2005–06.

(B) Kitusjärvi is at a higher latitude than Lake Monona and typically had more days of ice per winter than Lake Monona did.

(C) Kitusjärvi is at a higher latitude than Lake Monona and typically had fewer days of ice per winter than Lake Monona did.

(D) Lake Baikal had fewer days of ice in the winter of 1980–81 than it did in the winter of 2005–06.

The state of California has designated the southern water snake as an invasive species that could outcompete some of the state's native species. Many other states draw similar distinctions between invasive and native species. But researchers Alejandro Camacho and Jason McLachlan have pointed out that Earth's climate is changing in ways that challenge such designations. Climate changes may cause animals to leave their current ranges and establish new ones. Climate changes may also create good habitats in areas where a species couldn't live previously. These observations suggest that _____

15 Mark for Review

Which choice most logically completes the text?

(A) states such as California may need to reevaluate their classifications of species.

(B) it's useful at present for California to distinguish between invasive and native species in some instances but not in the case of the southern water snake.

(C) California was previously home to some southern water snakes but they were outcompeted by invading species.

(D) California should coordinate with other states to protect their native species from invasive species.

@DSATUZ

A recipient of the Wolf Prize in Chemistry and the Robert A. Welch Award, _____ known as femtochemistry when he devised an optical spectroscopy technique that allowed researchers to study ultrafast chemical reactions occurring within molecules.

17 Mark for

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) chemist Ahmed Zewail expanded the limits of measurability and introduced a new scientific discipline

(B) the result of chemist Ahmed Zewail's expansion of the limits of measurability was the introduction of a new scientific discipline

(C) the limits of measurability were expanded by chemist Ahmed Zewail who introduced a new scientific discipline

(D) the introduction of a new scientific discipline by chemist Ahmed Zewail led to the expansion of the limits of measurability

@DSATUZ

"Coalition" is the term for a group of male lions, and across the Greater Kruger National Park in South Africa, many coalitions vie for territory. Between 2010 and 2012, the lions of the Matimba _____ including Whiteface, Hairy Belly, and Ginger—thrived within a part of the park called the Manyeleti Game Reserve.

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) coalition—

(B) coalition

(C) coalition,

(D) coalition;

@DSATuz

Several advantages—the ability to react strongly with chip components, to avoid interference from other waves, and to be confined within tiny circuits—_____ acoustic waves as a promising alternative to electrical waves for transmitting data on computer chips; as a result, researchers are invested in developing more acoustic wave–based chips.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) having positioned

(B) have positioned

(C) by positioning

(D) positioning

@DSATUZ

At 2,200 years old, Bennett Juniper, a western juniper (*Juniperus occidentalis*) located in the United States, is one of the oldest known trees in the world. With over two millennia of climate data in its tree rings, a single tree like _____ claims dendrochronologist Valerie Trouet, can tell the history of the world.

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) this:

(B) this;

(C) this,

(D) this

@DSAT.uz

Like much scientific research, a noted 2021 study of northern elephant seals published in *Science Advances* reflects the combined efforts and expertise of multiple _____ Akinori Takahashi is listed first, Daniel Costa, Patrick Robinson, and other coauthors also made important contributions to the study.

21 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) researchers;

(B) researchers, while

(C) researchers: while

(D) researchers.

@DSATUZ

While researching a topic, a student has taken the following notes:

- Cheryll is the nickname of a dinosaur fossil specimen from the Late Cretaceous period.
- The Late Cretaceous period ended more than 65 million years ago.
- Cheryll is a member of the genus *Triceratops*.
- Cheryll is housed at the Palm Beach Museum of Natural History.
- The Palm Beach Museum of Natural History is in Wellington, Florida.

23 Mark for Review

The student wants to identify the institution in which Cheryll is housed. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Cheryll is housed at the Palm Beach Museum of Natural History in Wellington, Florida.

(B) Cheryll is the nickname of a dinosaur fossil specimen belonging to the genus *Triceratops*.

(C) Wellington, Florida, is home to a *Triceratops* fossil specimen nicknamed Cheryll.

(D) The *Triceratops* fossil specimen Cheryll lived more than 65 million years ago, in the Late Cretaceous period.

@DSATUZ

While researching a topic, a student has taken the following notes:

- A supercontinent is a single landmass made up of most or all of Earth's continents.
- Over time, continents merge together to form supercontinents, which then break apart.
- This process is believed to take hundreds of millions of years and is known as the supercontinent cycle.
- Columbia and Gondwana were supercontinents.
- Columbia formed about 1.8 billion years ago.
- Gondwana formed about 510 million years ago.

24 Mark for Review

The student wants to correct a misunderstanding that Gondwana formed before Columbia. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Columbia formed about 1.8 billion years ago, long before Gondwana's formation about 510 million years ago.

(B) Forming and breaking apart over hundreds of millions of years, supercontinents are made up of most or all of Earth's continents.

(C) Gondwana and Columbia formed at different times in Earth's history.

(D) Columbia and Gondwana were both supercontinents, single landmasses made up of most or all of the Earth's continents.

@DSATUZ

While researching a topic, a student has taken the following notes:

- A merchant ship is any ship hired to carry cargo or passengers.
- A ship's carrying capacity is also known as its deadweight tonnage (DWT).
- In 2021, there were a total of 392 merchant ships registered in Saudi Arabia.
- The combined DWT of these ships was 13.7 million tons.

25 Mark for Review

The student wants to identify the combined carrying capacity of Saudi Arabia's merchant ships in 2021. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) In 2021, Saudi Arabia had 392 merchant ships hired to carry cargo or passengers.

(B) The carrying capacity of a merchant ship is also known as its deadweight tonnage.

(C) The combined carrying capacity of Saudi Arabia's merchant ships was 13.7 million tons in 2021.

(D) There were a total of 392 merchant ships (ships hired to carry cargo or passengers) registered in Saudi Arabia in 2021.

@DSATuz

While researching a topic, a student has taken the following notes:

- The United States has designated more than 500 areas National Wildlife Refuges (NWRs).
- Some NWRs were established specifically to protect endangered species.
- The Crocodile Lake NWR is a 6,686-acre area in Florida, on the Atlantic coast.
- It was established to protect the endangered American crocodile.
- The Ellicott Slough NWR is a 139-acre area in California, on the Pacific coast.
- It was established to protect the endangered Santa Cruz long-toed salamander.

26 Mark for Review

The student wants to contrast the two NWRs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) While the Crocodile Lake NWR is in Florida, on the Atlantic coast, the Ellicott Slough NWR is in California, on the Pacific coast.

(B) Some NWRs were established specifically to protect endangered species such as the Santa Cruz long-toed salamander.

(C) Both the Crocodile Lake NWR and the Ellicott Slough NWR were established to protect endangered species.

(D) The US has designated more than 500 areas NWRs, including the Crocodile Lake NWR in Florida.

@DSATuz

While researching a topic, a student has taken the following notes:

- The French Republican calendar was developed during the French Revolution.
- It had twelve months divided into three weeks of ten days each, followed by five supplemental days.
- Each day was named after a plant, animal, agricultural tool, or mineral as a way of honoring an aspect of rural life.
- Frimaire is the third month of the French Republican calendar.
- The fifteenth day of Frimaire is named Chevreuil, which means “roe deer.”

27 Mark for Review

The student wants to explain why a day in the French Republican calendar’s month of Frimaire was named Chevreuil. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Developed during the French Revolution, the French Republican calendar has twelve months, including Frimaire.

(B) The month of Frimaire was divided into three weeks of ten days each, and one of those days was named Chevreuil.

(C) When the French Republican calendar was developed during the French Revolution, the fifteenth of Frimaire was named Chevreuil.

(D) Each day of the French Republican calendar year honors an aspect of rural life, which is why the fifteenth of Frimaire is named after an animal, the roe deer.

@DSATuz

To coders like Black Girls Code founder Kimberly Bryant, the computer programming languages PHP and S-Lang have a clear difference: the former is an object-oriented language while the latter is not. Because both languages require highly technical knowledge to understand, though, laypeople _____ to distinguish between them.

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) struggle

(B) were struggling

(C) had struggled

(D) struggled

@DSATuz

Although Oscar Micheaux's 1920 film *Within Our Gates* and Ossie Davis's 1970 film *Cotton Comes to Harlem* may have had some _____ at the time of their initial release, their place in critics' estimations is now more secure. In 2018, for example, critics for the *New York Times* described the former as "stunning" and the latter as "especially pointed."

1 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) precursors

(B) collaborators

(C) imitators

(D) detractors

@DSATUZ

Political scientists have found that although voters claim to prefer candidates who have nuanced perspectives on issues and who show a willingness to compromise, when asked to compare speeches expressing such views with speeches expressing _____ views, voters tend to regard the unyielding rhetoric of the latter more favorably.

2 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) dogmatic

(B) disingenuous

(C) vacillating

(D) banal

@DSATUZ

A number of recording artists _____ distinctions between Indigenous music and other musical styles. For example, Choctaw singer-songwriter Samantha Crain incorporated elements of Choctaw hymns into folk-pop on her single "When We Remain," and Oji-Cree/Mi'kmaq musician Anachnid combined Indigenous flute and powwow vocals with electronic music on her album *Dreamweaver*.

3 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) disguise

(B) reject

(C) observe

(D) replace

@DSATUZ

Today composer Scott Joplin is mainly celebrated for his catchy ragtime pieces “Maple Leaf Rag” and “The Entertainer.” However, by overlooking his less famous works, listeners will miss the full range of Joplin’s creativity. For instance, his waltz “Pleasant Moments” and his opera *Treemonisha* skillfully blend ragtime and classical music. These masterpieces deserve as much fame as Joplin’s biggest hits.

4 Mark for Review

Which choice best states the main purpose of the text?

(A) To describe the similarities and differences between ragtime music and opera

(B) To encourage music lovers to listen to music by many different composers

(C) To explain how Joplin learned to compose and perform ragtime music

(D) To argue that more attention should be given to Joplin’s lesser-known works

@DSATUZ

Desolation Island, first published in 1978, is a novel in Patrick O'Brian's Aubrey/Maturin series, which includes twenty completed books. Some critics have found fault with the abrupt endings of *Desolation Island* and other books in the series, saying that they do not finish conclusively but arbitrarily stop. But other critics argue that the books should not be thought of as discrete texts with traditional beginnings and endings but as a single incredibly long work, similar to other multivolume stories, such as John Galsworthy's *The Forsyte Saga*.

5 Mark for Review

Which choice best states the main purpose of the text?

(A) To explain why many critics find the Aubrey/Maturin novels to be remarkably entertaining despite flaws in the novels' structures

(B) To describe a characteristic of the Aubrey/Maturin novels and offer two differing viewpoints on this characteristic

(C) To argue that the Aubrey/Maturin series should have the literary renown of *The Forsyte Saga*

(D) To present a reason the unusual structure that O'Brian uses for *Desolation Island* makes it one of his most complex books

@DSATUZ

Text 1

Isaac Asimov, author of *Robots and Empire* and *I, Robot*, is highly regarded despite his mediocre writing style. His prose is workmanlike; his characters are flat and discuss ideas rather than emotions. That his work is enjoyable despite this is a testament to his prodigious imagination—even if people read his books only for the ideas, they will have plenty to consider.

Text 2

Asimov is critiqued for his style, but it is wrong to fault a writer for failing to do what he never intended to do. For example, although most of his novel *Foundation* consists of people discussing science and politics and we find out little about his characters' lives, Asimov wanted to convey the vast sweep of human history over centuries, and one of his points is that at such a timescale, individuals don't matter. Thus his lack of characterization is central to Asimov's thematic aims.

@DSATUZ

6 Mark for Review

Based on the texts, how would the author of Text 2 most likely respond to the evaluation of Asimov's writing style in the underlined claim in Text 1?

(A) By agreeing that Asimov's prose is flawed but disagreeing about the quality of his characters

(B) By arguing that Asimov's writing style should be judged against the styles of historians rather than those of fiction writers

(C) By asserting that Asimov's minimal characterization is likely a conscious choice that helps him convey an important idea

(D) By pointing out that the flatness of Asimov's characters is a feature of *Robots and Empire* and *I, Robot* but not of *Foundation*

Nicolas Dussex and colleagues relied on historical DNA (hDNA)—genomic data incidentally preserved in specimens housed in natural history collections—to investigate the evolutionary trajectory of the kākāpō parrot (*Strigops habroptilus*). Although this approach offers unique benefits, such as the ability to ascertain the biogeographical origins of organisms, it remains a relatively underutilized resource because hDNA is often to some extent degraded, a situation not easily remediable under current methodological paradigms and with extant DNA extraction and analysis technologies.

7 Mark for Review

[redacted] information in the text best supports which statement about hDNA?

(A) It may be underused because of its controversial status among scientists.

(B) It has thus far proved valuable mainly to researchers studying birds.

(C) It may yield insights that other types of genomic data cannot.

(D) It tends to be much more degraded than other types of DNA of comparable age.

@DSATUZ

Wenheng Zhang and colleagues have explored how convergent evolution—a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages—can result from a genetic mechanism shared by both lineages. Meanwhile, Bas J. Zwaan and colleagues have investigated how convergence occurs through different genetic mechanisms, but the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

8 Mark for Review

Which choice best states the main idea of the text?

(A) Green and Extavour's study addresses convergent evolution more comprehensively than the studies by Zhang and colleagues and Zwaan and colleagues do.

(B) Before Green and Extavour's study, convergent evolution was assumed to require a shared genetic mechanism between two lineages that share the trait.

(C) Green and Extavour's study was conducted using data from the studies by Zhang and colleagues and Zwaan and colleagues.

(D) Both the study by Zhang and colleagues and that by Green and Extavour compare convergence through shared genetic mechanisms to convergence through different genetic mechanisms.

@DSATUZ

Over time, marine mollusks called nautilids continually add segments to their shells, and the chemical composition of each segment is influenced by the available materials in the surrounding water when the segment forms. Paleontologist Amane Tajika and colleagues analyzed the septa (the walls between shell chambers) of one male (samples labeled "M") and one female (samples labeled "F") nautilid, finding that sample F02 had the same concentration of oxygen-18 as did water approximately 140 meters deep and that the oxygen-18 concentration in sample M23 matched that of water at approximately 385 meters deep. Tajika et al. concluded that sample F02 formed at a higher point in the water column than did M23.

9 Mark for Review

Which finding, if true, would most directly support the researchers' conclusion?

(A) Regardless of sex, nautilids spend most of their lives at depths of 50 meters or greater, where oxygen-18 concentrations typically differ from those higher in the water column.

(B) In nautilids, sex predicts septa growth rate more strongly than does oxygen-18 concentration in the surrounding water.

(C) Among nautilids, there is no effect of sex on the uptake of oxygen-18 from water or its deposition in septa.

(D) Male nautilids tend to show higher concentrations of oxygen-18 than do female nautilids inhabiting the same point in the water column.

@DSATUZ

Cane is a 1923 novel by Jean Toomer. In one portion of the novel, Toomer uses figurative language to connect the narrator's urban environment of Washington, DC, and the rural South of the narrator's past, writing, _____

10 Mark for Review

Which quotation from *Cane* most effectively illustrates the claim?

(A) "The Capitol dome looked like a gray ghost ship drifting in from sea."

(B) "The young trees had not outgrown their [planter] boxes then. V Street [in Washington, DC] was lined with them."

(C) "The [train] engines of this valley have a whistle, the echoes of which sound like iterated gasps and sobs. I always think of them as crude music."

(D) "And when the wind is from the South, soil of my homeland falls like a fertile shower upon the lean streets of [Washington, DC]."

@DSATUZ

Callie W. Babbitt, Hema Madaka, and colleagues assembled a database of materials used in consumer electronics by studying products in the lab and by gathering data from similar product studies. The team gave each of these studies a rating for level of traceability (with a higher rating for clearer descriptions of procedures) and for category consistency (with a higher rating for using materials categories more closely aligned with the categories in the team's database). Based on these ratings, a second research team concluded that the methodology was better explained in a study by Lee Ching-Hwa and Hsi Chi-Shiung than it was in a study by Oguchi Masahiro and colleagues.

11 Mark for Review

Which finding, if true, would most directly challenge the second research team's conclusion?

(A) The study by Lee and Hsi had a high consistency rating and a medium traceability rating.

(B) The study by Lee and Hsi had a lower consistency rating than the study by Oguchi and colleagues did.

(C) The study by Lee and Hsi had a lower traceability rating than the study by Oguchi and colleagues did.

(D) The study by Oguchi and colleagues had a low consistency rating and a low traceability rating.

@DSAT.UZ

Soil thawing in Alaska has been accelerating in response to climate warming, potentially enabling increased carbon dioxide (CO_2) absorption through greater productivity of arctic bramble (*Rubus arcticus*) plants and other vegetation, but also potentially enabling increased CO_2 output through greater heterotrophic respiration (CO_2 generated by the activity of soil microorganisms). Hydrologist Yonghong Yi and her colleagues developed a model incorporating numerous inputs—years of soil temperature and snow cover data among them—to evaluate the effects of warming on the CO_2 balance in Alaska, concluding that net CO_2 is likely to increase if warming hastens spring snow melt.

12 Mark for Review

Which finding, if true, would most directly support the researchers' conclusion?

(A) Early snow melt amplifies the effect of soil temperature on the growth of plant species such as *R. arcticus* and on the rate of heterotrophic respiration.

(B) Early snow melt reduces the amount of soil moisture available for the growth of plant species such as *R. arcticus*, and it raises the rate of heterotrophic respiration.

(C) Early snow melt slows the growth of plant species such as *R. arcticus* by reducing soil insulation, and it suppresses heterotrophic respiration.

(D) Early snow melt extends the period in which plant species such as *R. arcticus* can absorb CO_2 but has no effect on the rate of heterotrophic respiration.

@DSATUZ

The Canadian Longitudinal Study on Aging (CLSA-ÉLCV) is a longitudinal study surveying approximately 1,000 individuals in Canada to glean extended trends in aging. To carry out the necessary sampling of the population over many years, CLSA-ÉLCV needs extensive financial support, but this method provides valuable insights into causal relationships. However, when questions of causation are irrelevant, as with a public opinion study seeking only to reveal the proportion of a city's population that supports a particular tax proposal, longitudinal methods are unnecessary, and so _____

13 Mark for Review

Which choice most logically completes the text?

(A) conclusions drawn from CLSA-ÉLCV are likely to be more authoritative than those from the public opinion study.

(B) the expense of CLSA-ÉLCV is likely greater than the cost of longitudinal studies of public opinion.

(C) the success of the public opinion study likely requires significantly less financial outlay than that needed for CLSA-ÉLCV.

(D) longitudinal methods are suitable for studies of aging but ought to be avoided for those of public opinion.

@DSA Tuz

The many editions of James Joyce's 1922 novel *Ulysses* are not textually identical, and scholars debate which versions reflect Joyce's authorial intent. One no longer widely read edition is the 1984 "critical and synoptic edition" edited by Hans Walter Gabler, which followed French and German editorial theories rather than editorial traditions of the United States and United Kingdom and which was later found to have introduced errors due to Gabler's choice to consult facsimile manuscripts rather than using only originals. However, few Joyce scholars worldwide had expertise in such textual issues, and most of those who did worked on the edition with Gabler. So, it is unsurprising that initial scholarly reviews of the 1984 edition were mostly _____

14 Mark for Review

Which choice most logically completes the text?

(A) negative, since those Joyce scholars with the necessary expertise to write a review of the 1984 edition would be aware that facsimile manuscripts cannot be produced with a high enough fidelity to the original to ensure that relying on them will not introduce editorial errors.

(B) positive, since scholars who reviewed the 1984 edition were unaffiliated with its production and were mostly either Joyce specialists who were largely unfamiliar with editorial theories and practices or specialists in such theories and practices who were insufficiently familiar with Joyce.

(C) negative, since any scholar with expertise in editorial theories of the United States and United Kingdom as well as French and German editorial theories most likely worked with Gabler on the 1984 edition and would therefore not review it.

(D) positive, since *Ulysses* is a novel in English and the 1984 edition would therefore be more widely reviewed in United States and United Kingdom publications than in French and German publications.

@DSATuz

In Puerto Rico, it's not unusual for a city or town to be known by a nickname that corresponds to one of its notable features, like landscape, climate, famous residents, or chief exports. For example, the Puerto Rican municipality of Hatillo has also been called "the Land of Green Fields," a nickname that alludes to what the area is well _____ the lush greenery that surrounds it.

15 Mark for R...

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) known for:

(B) known for;

(C) known for

(D) known: for

@DSATUZ

A region of the Soviet Union encompassing the present-day city of Gorno-Altaysk, Russia, the Altai-speaking Gorno-Altai Autonomous Soviet Socialist Republic was in fact only nominally autonomous. President Mikhail Gorbachev's reforms, by helping the Gorno-Altai Republic and other Soviet republics established along ethnolinguistic lines assert their prerogatives, _____ to the Soviet Union's breakup.

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) contributing

(B) which contributed

(C) contributed

(D) having contributed

@DSATuz

Minerals can be classified by how much light passes through them. For example, all (or almost all) light passes through krotite, which is classified as _____ only some light to pass through, beryl is instead classified as translucent.

17 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) transparent, allowing

(B) transparent and allowing

(C) transparent. Allowing

(D) transparent allowing

@DSATuz

The artist Antoine Watteau's painting *Ceres (Summer)*, symbolically _____ to portray the season of summer as a time of anticipation, is a highly regarded example of allegory.

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) was functioning

(B) functions

(C) functioning

(D) has functioned

@DSATUZ

The relationship between genomes and epigenomes reveals how cells with identical DNA develop different _____ whereas the genome in each cell contains a complete DNA sequence, the epigenome consists of chemical compounds that determine which traits in the sequence will be expressed.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) functions and,

(B) functions:

(C) functions,

(D) functions

@DSAT.UZ

That the geographic center of North America lay in the state of North Dakota was conceded by all _____ establishing its precise coordinates proved more divisive.

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) involved;

(B) involved:

(C) involved

(D) involved,

@DSATuz

Monument 2 is the name given to an 11.8-ton stone statue of a human head that was created during the ancient Olmec civilization. It was discovered in Tabasco, Mexico, at the La Venta archaeological site. _____ at the San Lorenzo site in the state of Veracruz, another colossal head statue called Monument 4 was discovered.

21

Which choice completes the text with the most logical transition?

(A) Specifically,

(B) Elsewhere,

(C) For example,

(D) As a result,

@DSATUZ

In the Southeast Asian country of Vietnam, the percentage of the population living in cities increased from 33.8% to 37.3% between 2015 and 2020; this upward trend in urbanization was seen throughout that part of the continent. _____ from 2015 to 2020, the overall percentage of the Southeast Asian population living in cities rose from 47.2% to 50%.

22 Mark for Review

Which choice completes the text with the most logical transition?

(A) Nevertheless,

(B) Indeed,

(C) In reality,

(D) Later,

@DSATUZ

The cimbalom is an instrument with great cultural significance to Hungary. _____ the string instrument has played such a prominent role in the country's musical history that many have referred to it as the national instrument of Hungary.

23 Mark for Review

Which choice completes the text with the most logical transition?

(A) By comparison,

(B) Indeed,

(C) Conversely,

(D) In conclusion,

@DSATUZ

Ugandan American professor Peter Nazareth believed that Elvis Presley's music is best understood not as a homogeneous collection but as an anthology (because Elvis showcased the contributions of a wide range of gospel, blues, and rock artists). _____ Nazareth entitled his college course on Elvis and his music, which focused on Elvis's many musical influences, "Elvis as Anthology."

24 Mark for Review

Which choice completes the text with the most logical transition?

(A) In sum,

(B) That is,

(C) In addition,

(D) To that end,

@DSATUZ

- A copyright prevents a book's contents from being reproduced (published) without permission from the copyright holder.
- When a book's copyright expires, the book enters the public domain.
- Public domain books can be legally reproduced by anyone.
- *The Gift of Black Folk* is an essay collection by W.E.B. DuBois.
- It was published in 1924 and entered the public domain on January 1, 2020 (ninety-six years later).

25 Mark for Review

Which choice most effectively uses information from the given sentences to explain why *The Gift of Black Folk* entered the public domain?

(A) *The Gift of Black Folk*, an essay collection by W.E.B. DuBois, is freely available to the public.

(B) Once *The Gift of Black Folk* entered the public domain, anyone could publish its contents.

(C) Before *The Gift of Black Folk* entered the public domain, only its copyright holder could give permission for others to reproduce its contents.

(D) The copyright to *The Gift of Black Folk* expired in 2020, causing the essay collection to enter the public domain.

@DSATUZ

While researching a topic, a student has taken the following notes:

- Huazalingo is a municipality in the state of Hidalgo, Mexico.
- Municipalities are governmental regions responsible for providing many public services to their residents.
- One service they provide is water treatment.
- Huazalingo's population was 12,766 in 2020.
- Hidalgo is divided into 84 municipalities.

26 Mark for Review

The student wants to provide a specific example of what municipalities provide. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) In 2020, the municipality of Huazalingo had a population of 12,766.

(B) Huazalingo is one of 84 municipalities in Hidalgo, Mexico.

(C) Water treatment is just one of the many public services that municipalities provide.

(D) The municipality Huazalingo in Hidalgo is responsible for providing many public services.

@DSATUZ

While researching a topic, a student has taken the following notes:

- The Mohs scale of mineral hardness is a ten-point scale that orders minerals by hardness based on their ability to scratch other minerals.
- Minerals with larger numbers are harder than minerals with smaller numbers and can leave visible scratches on them.
- Minerals with smaller numbers are softer than minerals with larger numbers and cannot leave visible scratches on them.
- The mineral gypsum has a Mohs scale number of 2.
- The mineral fluorite has a Mohs scale number of 4.
- The mineral orthoclase has a Mohs scale number of 6.

27 Mark for Review

The student wants to compare the hardness of gypsum and orthoclase. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Orthoclase can leave visible scratches on fluorite, which is why orthoclase has a higher number than fluorite on the Mohs scale of mineral hardness.

(B) Orthoclase can leave visible scratches on gypsum, which means that orthoclase is harder than gypsum.

(C) The Mohs scale of mineral hardness can be used to order fluorite, gypsum, and orthoclase by their ability to scratch other minerals.

(D) A mineral with a Mohs number of 4, like fluorite, is harder than one with a Mohs number of 2, like gypsum.

@DSATUZ

The function g is defined by $g(x) = -7x + 40$. What is the value of $g(x)$ when $x = -2$?

(A) 54

(B) 31

(C) 26

(D) $\frac{38}{7}$

SODSAT

Each face of a fair 12-sided die is labeled with a number from 1 through 12, with a different number appearing on each face. If the die is rolled one time, what is the probability of rolling a 3?

A $\frac{1}{12}$

B $\frac{3}{12}$

C $\frac{9}{12}$

D $\frac{11}{12}$

@DSATUZ

$$8x + 14 = 8x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k?

14

/2)

Answer Preview: 14

@DSATUZ

imal

n (7/2)

A triathlon is a multisport race consisting of three different consecutive events. A triathlon participant plans to complete the cycling event with an average speed of 32,736 yards per hour. What is this average speed, in miles per hour? (1 mile = 1,760 yards)

18.6

Answer Preview: 18.6

@DSATUZ

Dan bought some apples and oranges. The apples cost $0.79 per pound, and the oranges cost $1.09 per pound. If Dan spent $5.94 in total and bought twice as many pounds of oranges as pounds of apples, how many pounds of oranges did Dan buy?

(A) 2

(B) 2.5

(C) 2.97

(D) 4

SODSAT

Triangle R has an area of 40 square centimeters (cm^2). Square S has side lengths of 6 cm. What is the total area of triangle R and square S, in cm^2?

(A) 23

(B) 26

(C) 46

(D) 76

@DSATUZ

$$y = 4x + 7$$

$$y = 3x + 9$$

The solution to the given system of equations is (x, y). What is the value of x?

(A) 2

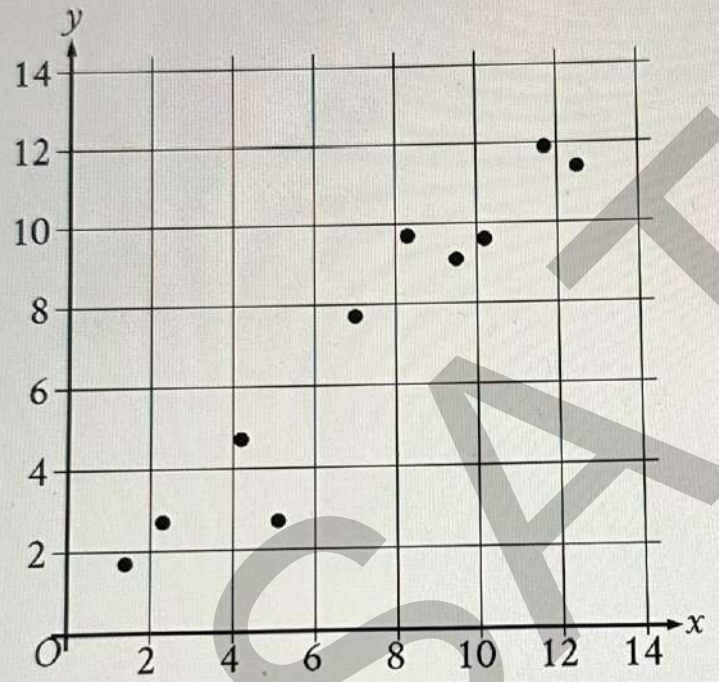
(B) 7

(C) 9

(D) 16

@DSATUZ

The scatterplot shows the relationship between two variables, x and y.



Which of the following equations is the most appropriate linear model for the data shown?

(A) $y = -2x$

(B) $y = -x$

(C) $y = x$

@DSATUZ

A rectangle has a length that is 17 times its width. The function $y = (17w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $17w$ in this context?

(A) The length of the rectangle, in feet

(B) The area of the rectangle, in square feet

(C) The difference between the length and the width of the rectangle, in feet

(D) The width of the rectangle, in feet

@DSATUZ

If $\frac{2a}{b} = 5.8$ and $\frac{a}{bn} = 23.2$, what is the value of $\frac{1}{n}$?

8

Answer Preview: 8

@DSATUZ

$$f(x) = 1{,}100(0.5)^{\frac{x}{15}}$$

The function f models the intensity of a beam, $f(x)$, in number of photons in the beam, x millimeters below the surface of a certain material. According to the model, what is the number of photons in the beam when it is at the surface of the material?

(A) 2

(B) 11

(C) 15

(D) 1,100

SODSAT

The gas mileage of a certain car is 25 miles per gallon when driven on the highway. If the car begins with 5 gallons of gas in the tank, which function predicts the amount of gas remaining in the tank, $f(d)$, after the car is driven d miles on the highway?

(A) $f(d) = \frac{d}{25} - 5$

(B) $f(d) = d - 5$

(C) $f(d) = 5 - \frac{d}{25}$

(D) $f(d) = 5 - d$

@DSATUZ

$$PC = N(85 - C)$$

The given equation relates the variables P, N, and C. Which expression represents the value of C for distinct positive values of P and N?

A $\frac{C(N+P)}{N}$

B $\frac{PC}{85-C}$

C $\frac{85N}{N+P}$

D $\frac{85N}{P+1}$

@DSATUZ

In triangle DEF, the measure of angle D is 36° and the measure of angle E is 101°. In triangle RST, the measure of angle R is 36° and the measure of angle S is 101°. Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST?

(A) The measure of angle F

(B) The measure of angle T

(C) The measure of angle F and the measure of angle T

(D) No additional information is needed.

@DSATUZ

What is the slope of the graph of $y = \frac{1}{4}(15x + 14) + 3x$ in the xy-plane?

27/4

Answer Preview: $\frac{27}{4}$

@DSATUZ

$$y = 2x + 9$$

One of the two equations in a system of linear equations is given. The system has no solution. Which of the following could be the second equation in this system?

(A) $y = -2x + 18$

(B) $y = -\frac{1}{2}x + 9$

(C) $y = \frac{1}{2}x - 9$

(D) $y = 2x - 18$

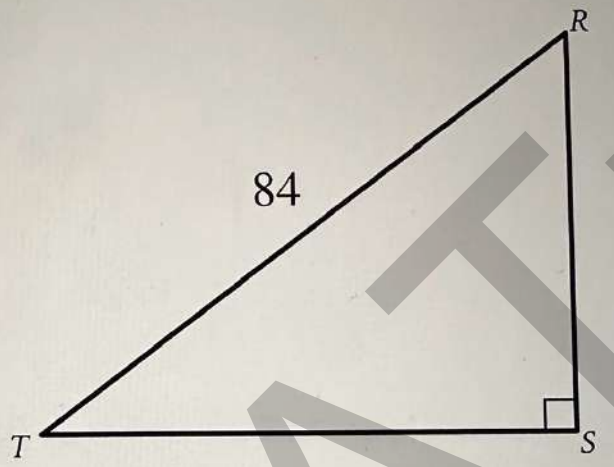
@DSATUZ

A linear model estimates the population of a city from 1995 to 2019. The model estimates the population was 56 thousand in 1995, 223 thousand in 2015, and x thousand in 2019. To the nearest whole number, what is the value of x?

256

Answer Preview: 256

DSAT



Note: Figure not drawn to scale.

In the triangle shown, $RS = \sqrt{167}$. What is the value of $\sin R$?

83/84

Answer Preview: $\frac{83}{84}$

@DSATuz

$$(x+4)^2+8(x+4)+16=0$$

How many distinct real solutions does the given equation have?

A Exactly one

B Exactly two

C Infinitely many

D Zero

@DSATUZ

1 Mark for Review

Line q in the xy-plane has a slope of $\frac{1}{4}$ and passes through the point $(0, 50)$. Which equation defines line q?

(A) $y = \frac{1}{4}x - 50$

(B) $y = \frac{1}{4}x - 46$

(C) $y = \frac{1}{4}x + 46$

(D) $y = \frac{1}{4}x + 50$

@DSATUZ

$$(x+2)^2+(y-18)^2=169$$

The graph of the given equation is a circle in the xy-plane. The point (a, b) lies on the circle. Which of the following is a possible value for a?

(A) -16

(B) -14

(C) 13

(D) 18

@DSATUZ

$$f(x) = 12(7)^x$$

The function f is defined by the given equation. If $g(x) = f(x + 3)$, which of the following equations defines the function g?

(A) $g(x) = 36(7)^x$

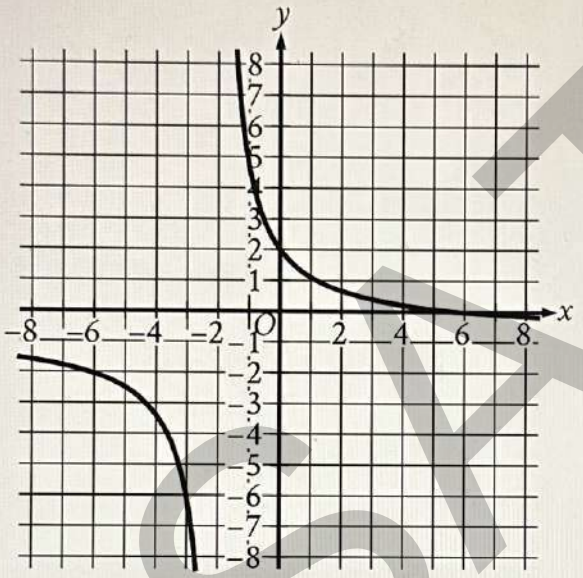
(B) $g(x) = 4{,}116(7)^x$

(C) $g(x) = 36(21)^x$

(D) $g(x) = 1{,}728(343)^x$

@DSATUZ

1 Mark for Review



The graph of $y = f(x)$ is shown in the xy-plane. What is the value of $f(0)$?

(A) -2

(B) $-\frac{7}{10}$

(C) 0

(D) 2

BDSAT UZ

2 Mark for Review

Which expression is equivalent to $(4nr^2 + 2nr) + (7n^2r + 3nr)$?

(A) $11n^2r^2 + 5nr$

(B) $11n^3r^3 + 5n^2r^2$

(C) $28n^3r^3 + 6n^2r^2$

(D) $4nr^2 + 7n^2r + 5nr$

SODSAT

The function g is defined by $g(x) = 3x$. For what value of x is $g(x) = 12$?

4

Answer Preview: 4

@DSATuz

4 Mark for Review

$$v = -\frac{w}{159x}$$

The given equation relates the distinct positive numbers v, w, and x. Which equation correctly expresses w in terms of v and x?

(A) $w = -159vx$

(B) $w = -\frac{159v}{x}$

(C) $w = -\frac{x}{159v}$

(D) $w = v + 159x$

@DSATUZ

5 Mark for Review

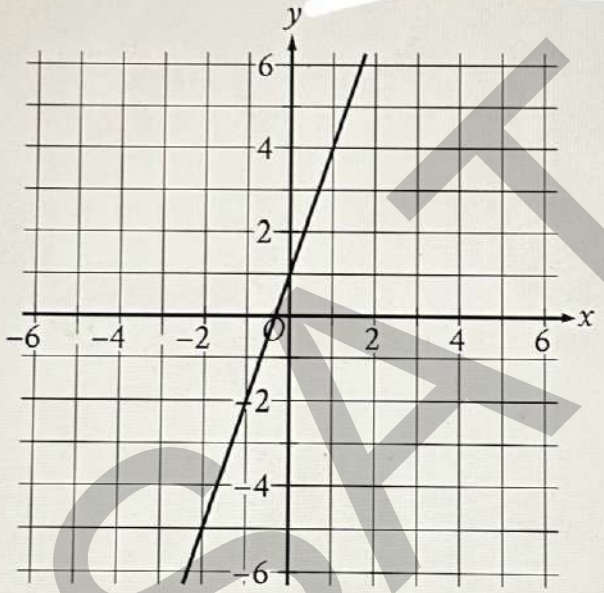
A real estate company offers a series of three webinars. 1,250 people attended the first webinar. 48% of the people who attended the first webinar attended the second webinar, and 38% of the people who attended the first and second webinars attended the third webinar. How many people attended all three webinars?

228

Answer Preview: 228

@DSATUX

6 Mark for Review



The graph of $y = f(x)$ is shown. What is the y-intercept of the graph of $y = f(x) + 16$?

(A) $(0, -15)$

(B) $(0, 17)$

(C) $(1, 16)$

(D) $(-16, 1)$

EDSAT.UZ

7 Mark for Review

The function f is defined by $f(x) = |x - 4x|$. What value of a satisfies $f(5) - f(a) = -6$?

(A) -11

(B) 2

(C) 7

(D) 18

@DSATUZ

e answer.
d up to 6 characters

l space, enter the decimal

space, enter it by

s an improper fraction (7/2)

ar sign.

otable: will eive credit
1/2 1/2
66 6

If $-9(13-5x)+3=-10(13-5x)+29$, what is the value of $13-5x$?

26

Answer Preview: 26

@DSATuz

9 Mark for Review

The minimum value of x is 9 less than 5 times another number n. Which inequality shows the possible values of x?

(A) $x \leq 5n - 9$

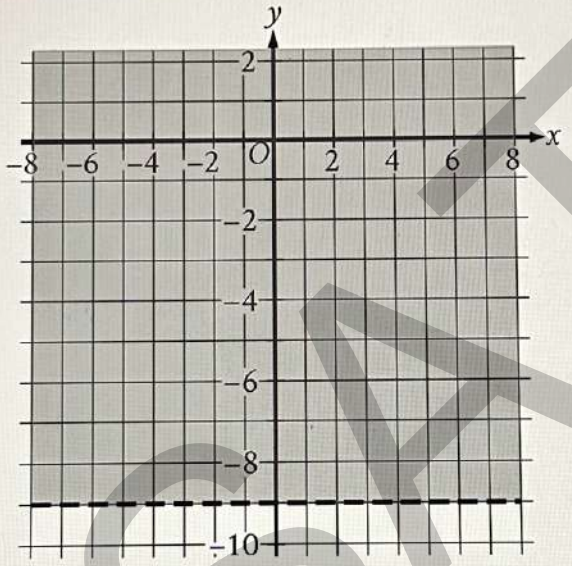
(B) $x \geq 5n - 9$

(C) $x \leq 9 - 5n$

(D) $x \geq 9 - 5n$

@DSATUZ

10 Mark for Review



The shaded region shown represents the solutions to the inequality $-14y < c$, where c is a constant. What is the value of c?

(A) 126

(B) 9

(C) −9

(D) −126

@DSATUZ

11 Mark for Review

At a constant temperature, a scuba diver combines two different concentrations of nitrox mix (oxygen-nitrogen gas mixture) to fill a tank to a total pressure of 160 atmospheres (atm), of which 34% is contributed by oxygen. Nitrox mix A contributes x atm of pressure to the tank, of which 26% is contributed by oxygen. Nitrox mix B contributes y atm of pressure to the tank, of which 42% is contributed by oxygen. Which system of equations represents this situation?

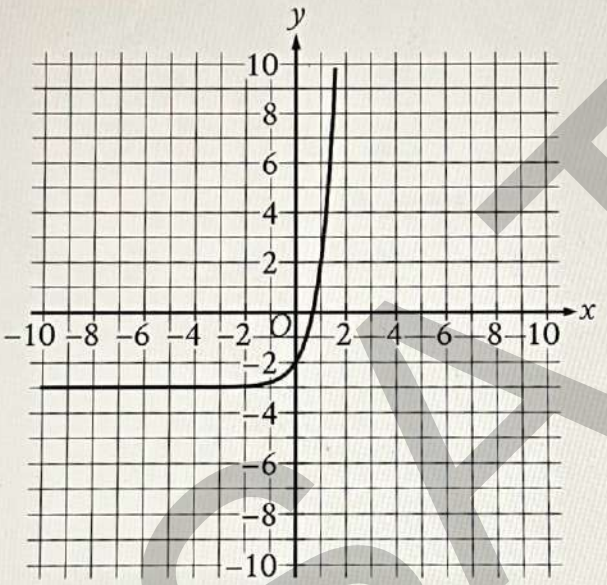
(A) $x + y = 160$
$0.26x + 0.42y = 0.34(160)$

(B) $x + y = 160$
$0.42x + 0.26y = 0.34(160)$

(C) $x + y = 0.34(160)$
$0.26x + 0.42y = 160$

(D) $x + y = 0.34(160)$
$0.42x + 0.26y = 160$

@DSATUZ



The graph of the equation $y = 5^x + k$ is shown, where k is a constant. What is the value of k ?

(A) -3

(B) -2

(C) 2

(D) 3

@DSATuz

13 Mark for Review

A certain town has an area of 3.59 square miles. What is the area, in square yards, of this town? (1 mile = 1,760 yards)

(A) 490

(B) 6,318

(C) 862,841

(D) 11,120,384

@DSATUZ

14 Mark for Review

The function f gives the product of a number, x, and a number that is 96 more than x. Which equation defines f?

(A) $f(x) = x^2 + x + 96$

(B) $f(x) = x^2 + 96$

(C) $f(x) = x^2 + 96x$

(D) $f(x) = x^2 + 96x + 96$

@DSATUZ

One of the factors of $3x^3 + 51x^2 + 216x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b?

8

Answer Preview: 8

@DSATUZ

x	$h(x)$
0	1.34
2	1.83
4	2.51

The table shows the exponential relationship between the number of years, x, since Hana started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

(A) $h(x) = 1.17(0.34)^x$

(B) $h(x) = 1.17(1.34)^x$

(C) $h(x) = 1.34(0.17)^x$

(D) $h(x) = 1.34(1.17)^x$

©DSATUZ

Kenny engages in up to 3 types of exercise each week for a total of 6 hours while training for a triathlon. Kenny swims for the same number of minutes each week. The equation $y = 360 - x - 160$ represents the situation where Kenny bikes for x minutes during a week and runs for any remaining training time y, in minutes. If this equation is graphed in the xy-plane, which statement is the best interpretation of the x-intercept of the graph?

(A) During a week when Kenny swims for 200 minutes, he bikes for 160 minutes.

(B) Each week, Kenny bikes and runs for a total of 200 minutes.

(C) During a week when Kenny doesn't bike, he runs for 200 minutes.

(D) During a week when Kenny doesn't run, he bikes for 200 minutes.

@DSATUX

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 140)^\circ$. Which of the following could NOT be the sum of the measures of any two of these angles?

A $(-18x + 280)^\circ$

B $(-18x + 640)^\circ$

C $(18x - 280)^\circ$

D 180°

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rs

decimal

ction (7/2)

$$\frac{5}{7}(5x+7)\left(x+\sqrt{5k+7}\right)\left(x-\sqrt{5k+7}\right)=0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 84. What is the value of k?

10.6

Answer Preview: 10.6

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Each of the following frequency tables represents a data set. Which of these frequency tables represents the data set with the smallest standard deviation?

(A)

Value	Frequency
5	0
6	10
7	30
8	10
9	0

(B)

Value	Frequency
5	6
6	12
7	14
8	12

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9	6

C

Value	Frequency
5	10
6	10
7	10
8	10
9	10

D

Value	Frequency
5	13
6	9
7	6
8	9
9	

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	Volume (cubic units)
Right circular cylinder A	500π
Right circular cylinder B	$32{,}000\pi$

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 10 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? (The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

4500

Answer Preview: 4500

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rs

decimal

tion (7/2)

Function f is a quadratic function. The graph of $y = f(x)$ in the xy-plane has a vertex at $(6, 10)$, contains the point $(5, 8)$, and has a y-intercept at $(0, a)$. The graph of $y = 4 \cdot f(x)$ has a y-intercept at $(0, b)$. What is the positive difference between a and b?

186

Answer Preview: 186

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MARCH 2024 Answers

MODULE 1 (Verbal)
1 D
2 D
3 C
4 C
5 C
6 A
7 D
8 A
9 B
10 D
11 D
12 B
13 C
14 D
15 A
16 A
17 A
18 A
19 B
20 C
21 C
22 –
23 A
24 A
25 C
26 A
27 D

MODULE 2(verbal)
1 D
2 A
3 C
4 D
5 B
6 C
7 B
8 A
9 C
10 D
11 C
12 B

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13 D
14 B
15 A
16 C
17 C
18 C
19 B
20 A
21 B
22 B
23 B
24 D
25 D
26 C
27 B

MODULE 1 (math)
1 D
2 A
3 a
4. 14
5. 18.6
6 D
7 D
8 A
9 C
10 A
11. 8
12 D
13 C
14 C
15 D
16. 27/4
17 D
18. 256
19. 83/84
20 A
21 B
22 B

MODULE 2(math)
1 D
2 D

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3. 4
4 A
5. 228
6 B
7 C
8. 26
9 B
10 A
11 A
12 A
13 D
14 C
15. 8
16 D
17 B
18 A
19. 10.6
20 A
21. 4500
22. 186

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